Movie Project Tech Report

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1 Introduction

For this project, we set out to explore relationships between bad movies. We were interested in discovering whether bad movies tend to be highly related, perhaps clustered in certain genres or with a recurring cast of actors or directors, or whether they tend to be more diffuse. Broadly speaking, our research question is to determine what factors poorly rated movies have in common and how strongly they are connected to one another.

To investigate this question, we used network analysis techniques to study structural relationships between films and the people involved in their production. Network analysis provides a natural framework for this type of question because it allows us to model relationships such as collaborations between actors, shared directors, or common genre classifications. By representing these relationships as graphs, we can use centrality measures, community detection, and degree distributions to better understand patterns that may not be obvious from individual movie data alone.

We constructed multiple networks to examine different possible explanations for low ratings. These include a director network to examine whether certain directors repeatedly produce poorly rated films, a genre network to determine whether certain genre combinations are more common among poorly rated movies, and a cast collaboration network to analyze whether groups of actors frequently appear together in these films. By comparing these different networks, we aim to better understand whether poorly rated movies are mostly isolated cases or whether they emerge from identifiable structural patterns.

Through these networks, we attempt to answer several related questions: Do certain individuals repeatedly appear in poorly rated films? Do certain genres tend to produce lower-rated movies? Do collaboration patterns among actors reveal small communities associated with poorly received films? By answering these questions, we hope to better understand whether poor reception is mostly random or whether there are identifiable structural factors associated with these movies.

2 Exploratory Data Analysis

We defined bad movies as those with an IMDB rating of 2/10 or lower, and filtered out those with fewer than 10 ratings, which reduced our dataset from more than 86,000 to a manageable 871 films.

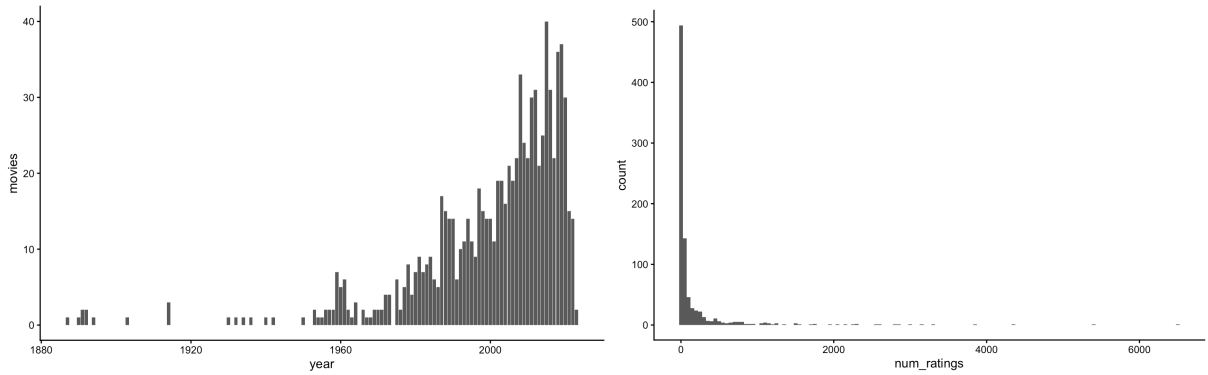


Figure 1: Time and number of ratings distributions

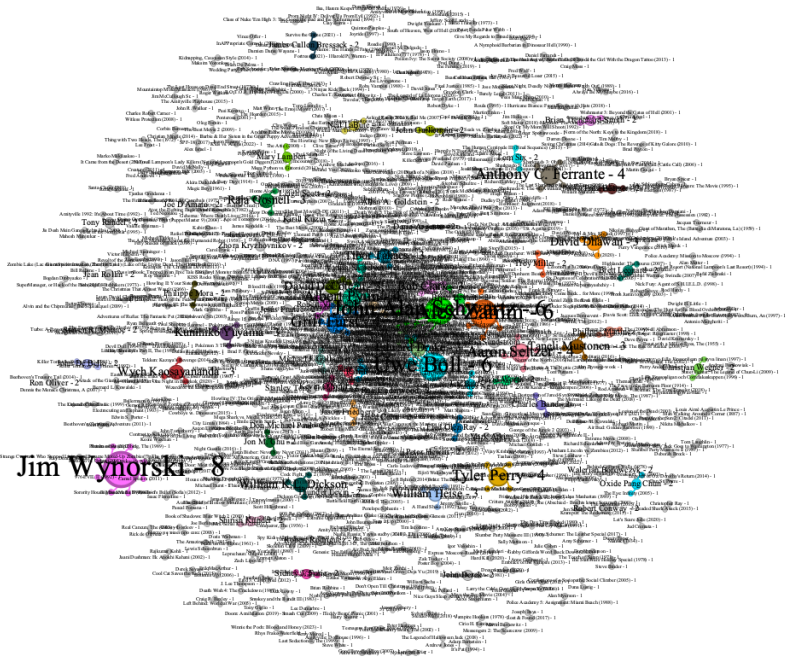
The dataset (perhaps unfairly) includes movies from as far back as 1887, but the large majority are from the past 50 years, which makes sense as these are more likely to have IMDB ratings. As far as number of ratings, there is clearly a very skewed distribution, with a mean of 177 and a median of just 21. However, as mentioned earlier, those with a truly tiny number of ratings were filtered out, so there must be at least some consensus that these movies truly are bad, rather than just a couple bad experiences.

3 Directors Network

To investigate whether certain directors are associated with poorly rated films, we constructed a bipartite network connecting directors and movies with IMDb ratings below 2.0. In this network, nodes represent directors and movies, and an undirected edge connects a director to a movie if they directed it. This structure represents a professional relationship between directors and their films and allows us to analyze whether extremely low-rated movies tend to be associated with particular individuals.

The final network contains 1541 nodes and 828 edges, with an average degree of approximately 1.075. The low average degree suggests that most directors in this dataset are associated with only one poorly rated film. This indicates that extremely low ratings are typically isolated outcomes rather than recurring results for most directors.

To identify important directors in this network, we examined degree centrality, PageRank, and betweenness centrality. Among these measures, degree centrality is the most meaningful here because it directly measures how many poorly rated movies a director is connected to. As shown below, Jim Wynorski has the highest degree with eight low-rated films, followed by several directors including Alex Zamm, John Alan Schwartz, and Uwe Boll with six each. This suggests that while most poorly rated films are isolated cases, a small number of directors repeatedly appear in this subset.


Jim Wynorski - 8
Alex Zamm - 6
Uwe Boll - 6
Anthony C. Ferrante - 4
Tyler Perry - 4
William Dickson - 2
William Hess - 2
Aaron Seltzer - 3
Jason Friedberg - 3
Ron Oliver - 2
Stanley Tong - 2
Joe D'Amato - 2
Rita Cheng - 2
Yoshihiro Nishimura - 2
Tom Holland - 2
Ivan Reitman - 2
Ron Howard - 2
Jon Moscow - 2
Ray - 2
Oxide Pang Chun - 2
Wych Kaosayananda - 2
Kunihiko Ikuyama - 2
Zhora Kryzhovnikov - 2
Tom Holland - 2
Dmitrii Kvitsinskii - 2
Dmitrii Kvitsinskii - 2
Dmitrii Kvitsinskii

	Label	Degree	PageRank	Betweenness.Centrality
1	Jim Wynorski	8	0.002734844	2.36e-05
2	Alex Zamm	6	0.002138868	1.27e-05
3	John Alan Schwartz	6	0.002138868	1.27e-05
4	Uwe Boll	6	0.002138868	1.27e-05
5	Aaron Seltzer	4	0.001542893	5.06e-06
6	Albert Pyun	4	0.001542893	5.06e-06
7	Anthony C. Ferrante	4	0.001542893	5.06e-06
8	Griff Furst	4	0.001542893	5.06e-06
9	Tyler Perry	4	0.001542893	5.06e-06
10	Bob Clark	3	0.001244905	2.53e-06

Figure 3: Table

Betweenness centrality values are generally very small throughout the network. This is expected because most directors only connect to their own films, meaning few nodes serve as bridges between otherwise separate parts of the network. This reinforces the observation that the network consists largely of small disconnected components rather than a highly interconnected structure.

To further understand the structure of the network, we examined the degree distribution of directors. The histogram below shows that most directors have degree 1, meaning they directed only one poorly rated movie, while only a small number directed multiple poorly rated films.

We also constructed a log-log complementary cumulative distribution function (CCDF) of the director degree distribution to examine whether the network exhibits heavy-tailed behavior. The CCDF plot shows a sharp decline after degree 1 followed by a sparse tail extending to degree 8. This indicates that while most directors are associated with only one poorly rated film, a very small number are associated with multiple such films.

Although the CCDF shows a long tail, the small number of high-degree directors suggests only limited evidence of a power-law distribution. Instead, the network appears to be dominated by many isolated cases with only a few directors repeatedly appearing among the lowest-rated movies. This suggests that extremely poor ratings are generally not concentrated among a large group of consistently poorly performing directors, but instead arise mostly from individual projects.

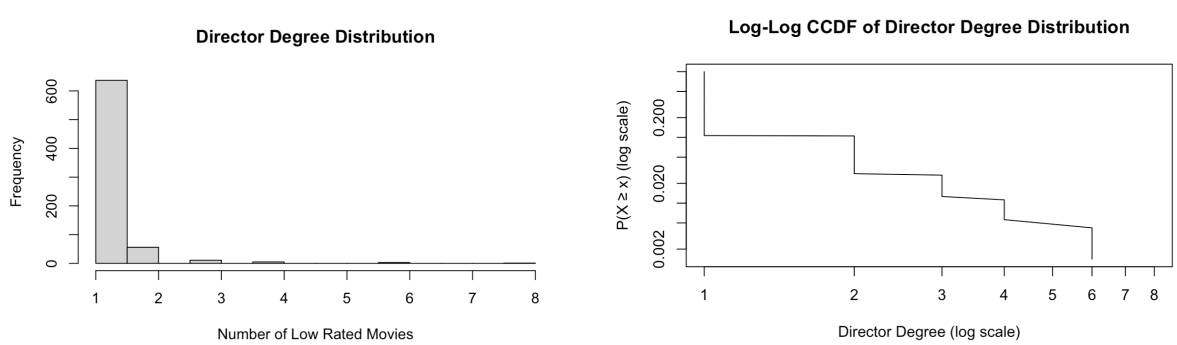


Figure 4: Histogram and log-log CCDF of director degree distribution

Modularity analysis produced a very high modularity value (approximately 0.998), indicating strong separation between components. This reflects the structure of the network, where most directors form

small disconnected groups with their films rather than belonging to large collaborative communities. In this network, communities therefore represent isolated director–movie relationships rather than meaningful clusters of collaboration.

Overall, the director network suggests that poorly rated films are usually isolated events rather than the result of consistent patterns among most directors. While a small number of directors appear repeatedly among the lowest-rated movies, the majority appear only once. This suggests that producing a poorly rated film does not necessarily indicate a persistent pattern of poor performance, but rather reflects isolated outcomes in most cases.

4 Genre Network

Another way we thought to analyze our data was to look at genre connections between bad movies. Initially, we connected movie nodes with an edge if they shared any genre, but this produced an extremely dense "spaghetti" graph, so we tightened the restriction to multiple shared genres, with edges weighted by number of shared genres. This also has the effect of locating movies that are more similar to each other, as simply being a comedy doesn't make films that similar, even if both those films received low ratings, but two bad romantic comedies are likely have quite a bit in common.

539 movies (62%) made the cut, which is telling as many bad movies will try to do much more than they should. Average degree is 72, and average weighted degree is 151, which makes sense because all edges have weight at least two per our setup. Both metrics are strongly correlated (r = 0.84) with number of genres, which makes sense as movies with more genres are more likely to have two or more genres in common with other movies, but there is still room for interesting behavior to emerge in which smaller combinations of genres can produce higher degree nodes than larger combinations.

title	genres	weighted degree	eigenvector	pagerank	closeness	betweenness
Hobgoblins	Action|Comedy|Horror|Sci-Fi|Thriller	699	1	0.006576	0.700528	0.049224
2012 Doomsday	Action|Adventure|Fantasy|Horror|Sci-Fi|Thriller	718	0.998629	0.006396	0.687824	0.04625
Alone in the Dark	Action|Horror|Sci-Fi|Thriller	533	0.890468	0.004838	0.62178	0.010644
Anaconda: The Offspring	Action|Horror|Sci-Fi|Thriller	533	0.890468	0.004838	0.62178	0.010644
House of the Dead 2	Action|Horror|Sci-Fi|Thriller	533	0.890468	0.004838	0.62178	0.010644
Killer Tomatoes Strike Back!	Action|Comedy|Horror|Sci-Fi	511	0.8466	0.004946	0.636691	0.016641
Birdemic 2: The Resurrection	Action|Comedy|Horror|Sci-Fi	511	0.8466	0.004946	0.636691	0.016641
Mega Piranha	Action|Adventure|Horror|Sci-Fi	457	0.822446	0.004262	0.605473	0.00763
Sharknado 4: The 4th Awakens	Action|Adventure|Horror|Sci-Fi	457	0.822446	0.004262	0.605473	0.00763
Kids vs. Aliens	Action|Adventure|Horror|Sci-Fi	457	0.822446	0.004262	0.605473	0.00763

Figure 5: Table 2

Indeed, we see that the two most important movies in the network have a dubious five and six genre tags respectively, but the rest of the top ten are just four, despite there being plenty of other nodes with five-plus genres tagged. Every film in the top ten is action, horror and sci-fi, which shows that these are the three main genres connecting bad movies. Horror and sci-fi are definitely pretty easy to do poorly, while action we suspect is just a very broad category in general. It's also interesting to note that eight of the top ten reference some kind of monster, from hobgoblins to snakes to sharks to piranhas to...killer tomatoes?

Figure 6: Sample posters

Turning to broader network behavior, the log-log plot does not show a power law relationship.

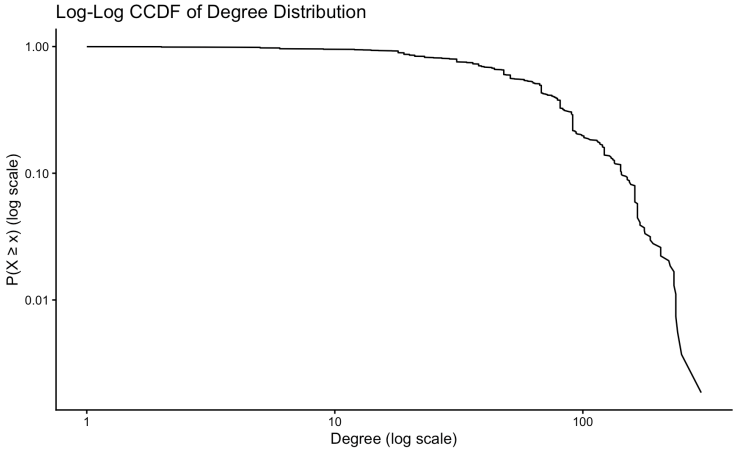


Figure 7: Log-log CCDF of genre degree distribution

Given the way our network is set up, this isn't too surprising. Since any pair of genres can form an edge, films with three genres have three possible combinations, and even films with just two, the bare minimum for inclusion in the network, could still have a really common pair, such as action/sci-fi. However, there is still a smaller class of the biggest hubs, with four, five or six genre tags, so we observe a sort of ballooning distribution, with few leaves or hubs but many mid-degree nodes.

Finally, here is the network itself:

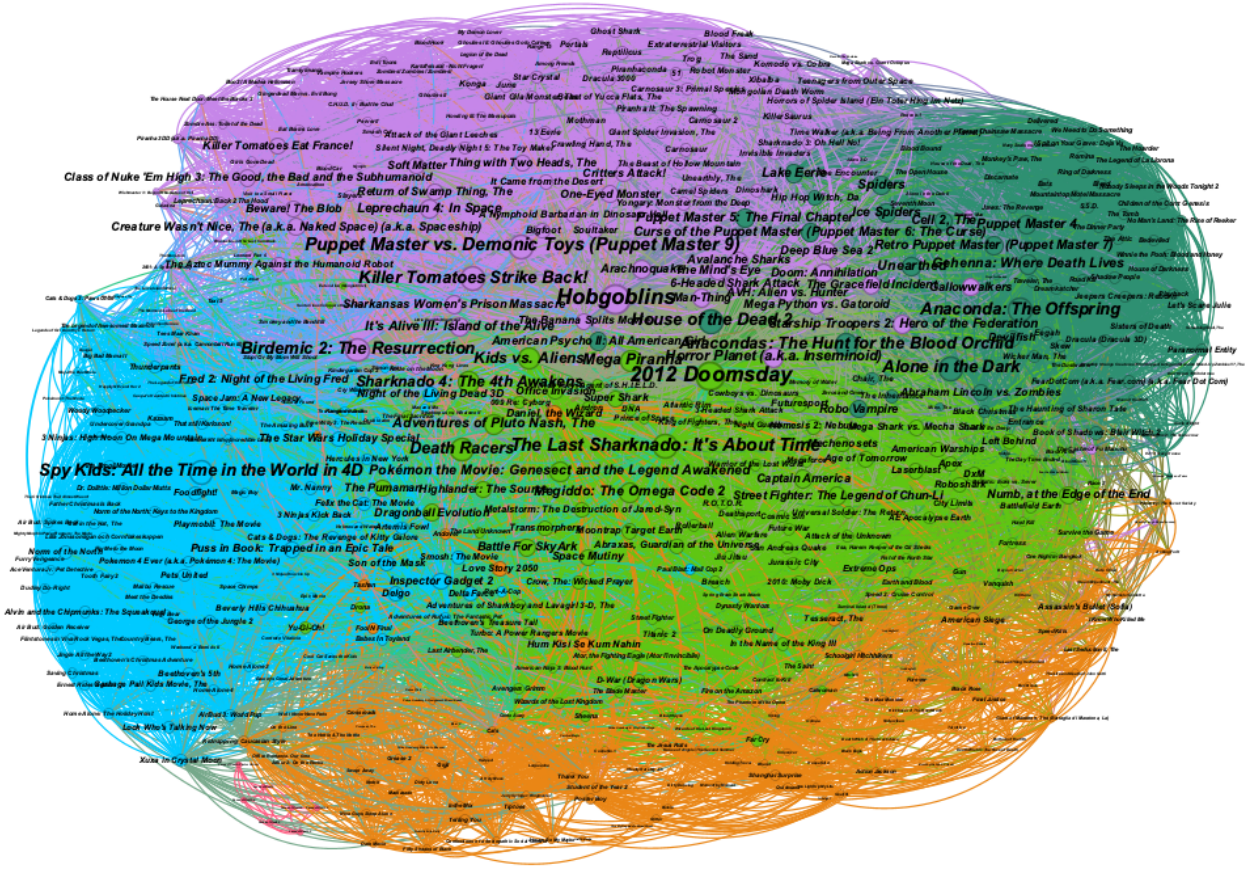


Figure 8: Genre network visualization

Gephi detected 6 communities at a robust range of resolutions, this graph coming with 1.0 and a modularity of 0.454. However, the sixth community is very hard to see in pink at the bottom left corner, consisting of seven movies from the *Faces of Death* series, the only documentary horror films in the dataset.

Putting that aside, we see five main communities in the network. It's not actually a very clear genre split, because all of these movies have multiple genres, but each does have a distinct feel. Starting in the top right with the dark green, this community clearly has a horror feel, with titles such as *House of the Dead 2*, *American Psycho II* and several installments in the nine-part *Puppet Master* series standing out (unsurprisingly, there are a lot of sequels in the dataset). Moving counterclockwise, we come to a similar horror-esque community, but here there's a distinct monster/otherworldly theme, highlighted by *Hobgoblins*, *The Creature Wasn't Nice* (which was also released as *Naked Space* and *Spaceship*), and *Birdemic 2* (which is definitely not a case of the sequel ruining the series). Moving down, the light green community has a more sci-fi feel, although there are still a lot of references to various creatures, another general theme in the dataset.

To the left, the blue community has a strong action/adventure feel, which includes some personal favorites in *Foodfight!* and the *Star Wars Holiday Special*. Finally, the orange community is clearly much more diffuse, but has a lot of romantic and comedy titles. Our hypothesis is that many of the other movies try to incorporate these elements, as opposed to specifically horror or action films, which spreads out these nodes, and also contributes to a notable lack of hubs in this community.

5 Cast Network

To further investigate whether poorly rated films share common structural patterns, we constructed an actor collaboration network using the top three billed actors from each movie with rating ≤ 2.0. In this network,

each node represents an actor, and an undirected edge connects two actors if they appeared together in the top three billed cast of the same low-rated movie. This network therefore represents professional collaboration relationships between actors.

We limited the network to the top three billed actors to focus on meaningful collaborations rather than incidental cast appearances. Including the full cast would introduce many weak connections from minor roles, while the top three actors better capture the primary collaborative structure of each film.

The final network contains 2149 actors and 2406 edges, with an average degree of approximately 2.239. Compared to the director network, this network shows a higher level of connectivity, suggesting that actors may appear repeatedly across poorly rated films more often than directors do. The network diameter is 9 and modularity is approximately 0.996, indicating strong separation into small collaboration groups.

To better illustrate the structure of the network, the visualization below shows the full actor network without labels. Removing labels improves readability and highlights the clustering structure and disconnected components present in the network.

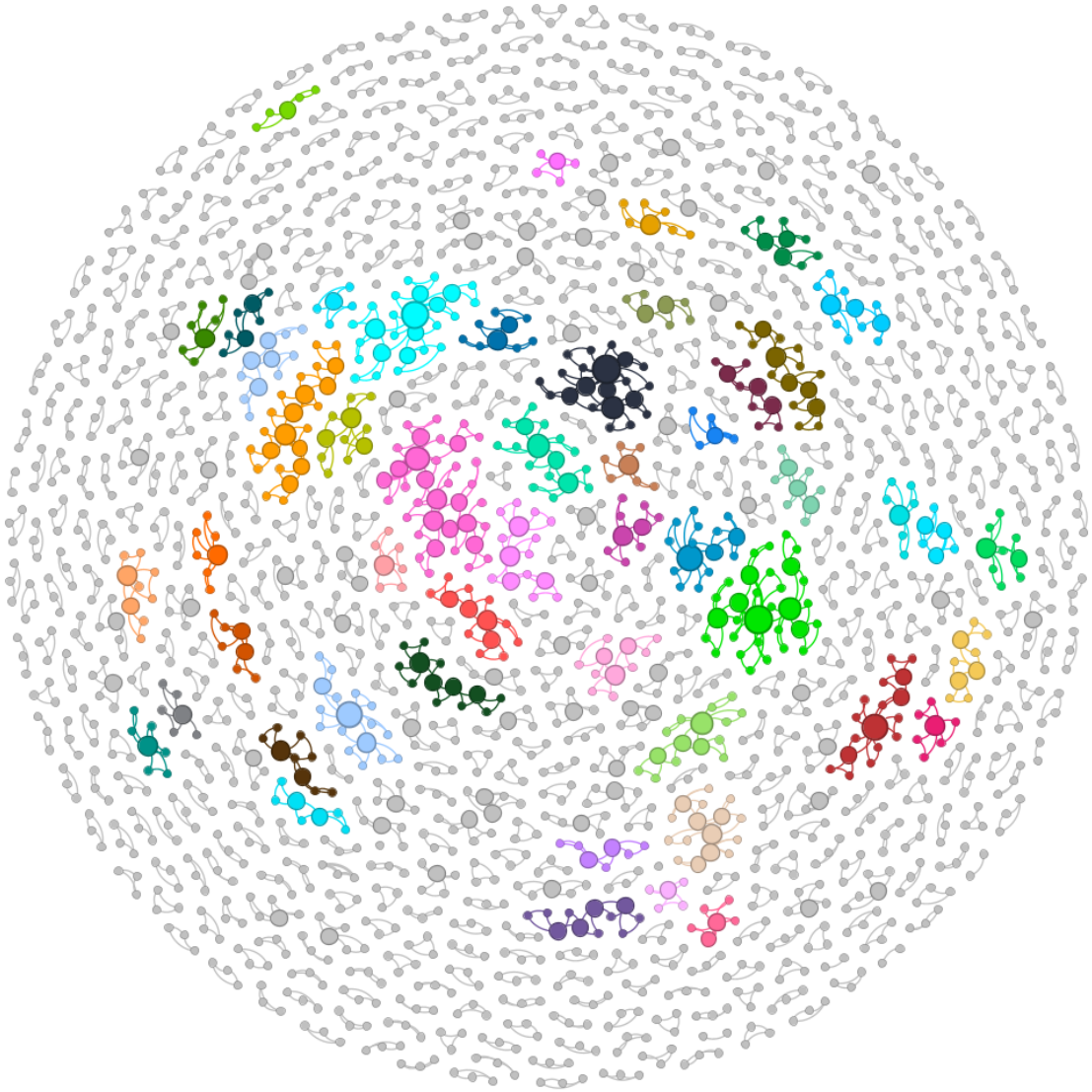


Figure 9: Actor collaboration network based on top three billed cast

Because labeling all nodes reduces readability, the next visualization shows a filtered version where only high-degree actors are labeled. This allows clearer identification of the most central actors while preserving the overall network structure. Selective labeling improves interpretability while maintaining the structural context of the network.

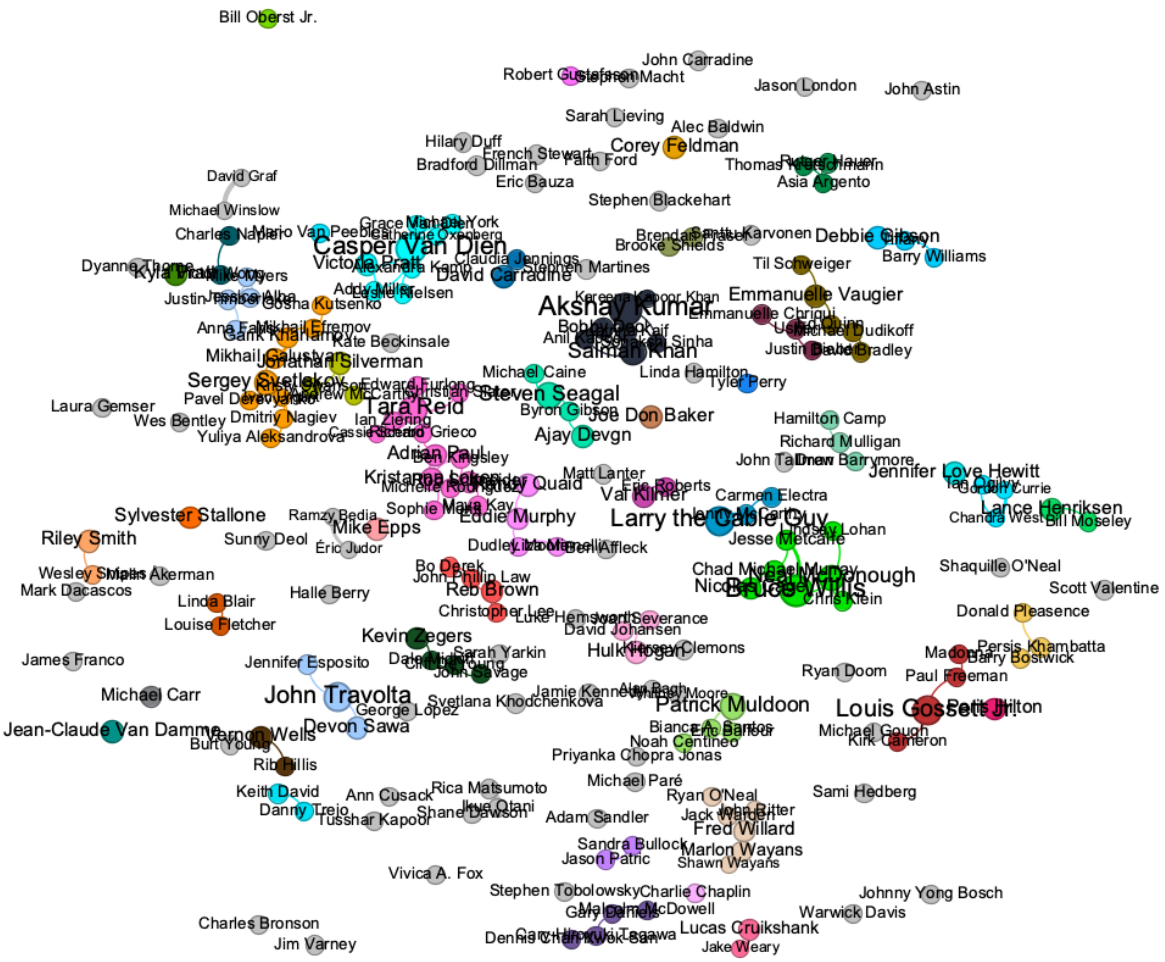


Figure 10: Filtered actor network showing only high-degree actors

To identify important actors within this network, we examined degree centrality, PageRank, and betweenness centrality. Degree centrality is particularly meaningful in this network because it directly measures how frequently an actor collaborates with others in poorly rated films. The most central actors include Bruce Willis and Akshay Kumar, each appearing in twelve collaborations within this subset. Other highly connected actors include Casper Van Dien, John Travolta, Louis Gossett Jr., Steven Seagal, and Tara Reid. These actors appear repeatedly across poorly rated films, suggesting that certain performers may be more likely to participate in projects that receive low audience ratings.

	Label	Degree	PageRank	Betweenness.Centrality
1	Bruce Willis	12	0.001664239	1.33e-04
2	Akshay Kumar	12	0.001578241	4.45e-05
3	Casper Van Dien	11	0.001530888	1.05e-04
4	Louis Gossett Jr.	10	0.001494709	4.16e-05
5	John Travolta	10	0.001496294	4.16e-05
6	Larry the Cable Guy	10	0.001524708	3.12e-05
7	Tara Reid	9	0.001241921	1.41e-04
8	Neal McDonough	8	0.001151454	1.01e-04
9	Salman Khan	8	0.001068544	2.62e-05
10	Steven Seagal	8	0.001205907	3.64e-05

Figure 11: Actors with highest degree centrality

The degree distribution further supports this observation. Most actors have very low degree values, typically appearing in only one or two poorly rated films, while only a small number of actors appear frequently. This produces a strongly right-skewed distribution typical of collaboration networks.

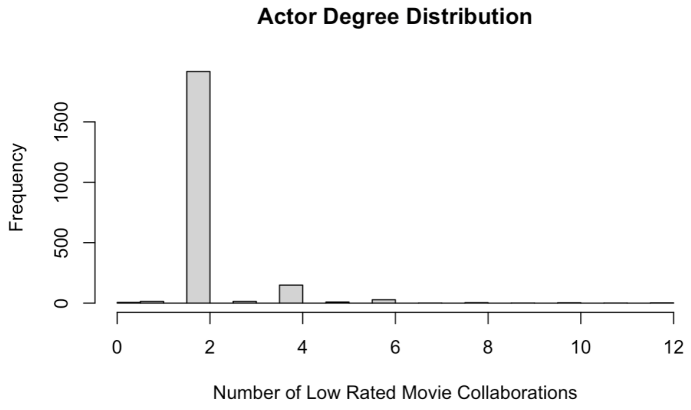


Figure 12: Actor degree distribution

To further analyze this pattern, we constructed a log-log complementary cumulative distribution function (CCDF) of the actor degree distribution. The CCDF shows a gradual decline with a visible long tail extending to degree 12. Compared to the director network, the tail is slightly heavier, suggesting stronger evidence of repeated appearances among certain actors.

Although the CCDF shows some heavy-tailed behavior, the relatively small number of high-degree actors suggests only limited evidence of a strict power-law distribution. Instead, the results suggest that while most actors appear only once or twice in poorly rated films, a small group appears repeatedly. This pattern may indicate that certain actors specialize in lower-budget or niche productions that tend to receive lower ratings.

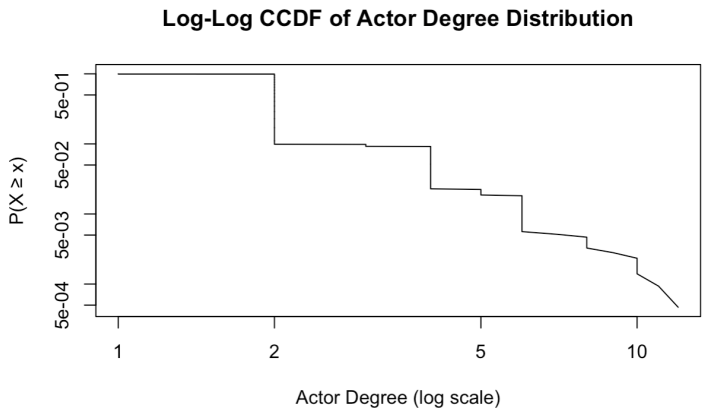


Figure 13: Log-log CCDF of actor degree distribution

Community detection produced a very high modularity score, indicating strong separation between collaboration groups. These communities largely correspond to actors who frequently collaborate within the same productions or production circles. This suggests that poorly rated films often emerge from small clusters of recurring collaborations rather than random actor combinations.

Overall, the cast network suggests that while most actors appear only occasionally in poorly rated films, a small number appear repeatedly and form collaboration clusters. Compared to the director network, which showed mostly isolated cases, the cast network provides stronger evidence that repeated professional relationships may play a role in the structure of poorly rated films.

6 Conclusion

In this project, we constructed multiple networks to better understand whether poorly rated movies share structural similarities. By examining director, genre, and cast collaboration networks, we were able to explore whether low ratings tend to be isolated events or whether they emerge from recurring patterns of collaboration.

The director network suggested that most poorly rated movies are isolated cases, with most directors only appearing once in the dataset. Only a small number of directors were connected through multiple poorly rated films, suggesting that directing low-rated movies is typically not a repeated pattern. The genre network showed that certain genres frequently co-occur in poorly rated movies, indicating that some genre combinations may be more prone to producing films that receive low audience ratings. Finally, the cast network showed stronger clustering patterns, suggesting that small groups of actors sometimes repeatedly collaborate in poorly rated productions.

Across all three networks, the degree distributions showed strong right-skewed behavior, where most nodes had very small degree and only a few nodes had larger connectivity. The CCDF plots showed some heavy-tailed behavior but did not provide strong evidence of a true power-law distribution. Instead, the results suggest that poorly rated films are mostly scattered cases with a few small clusters of repeated collaborations.

One limitation of this study is that we only considered structural relationships and did not incorporate other important variables such as budget, production company, or audience demographics. Additionally, restricting the cast network to the top three billed actors simplifies the network but may omit important supporting actor relationships. Future work could expand this analysis by incorporating additional movie attributes or by examining temporal patterns to see whether collaboration patterns change over time.

Overall, our results suggest that while poorly rated movies do not form one large connected structure, small collaboration clusters do exist, particularly among actors. These findings suggest that network analysis provides a useful way to understand structural patterns in the film industry, even among movies that are not critically successful.